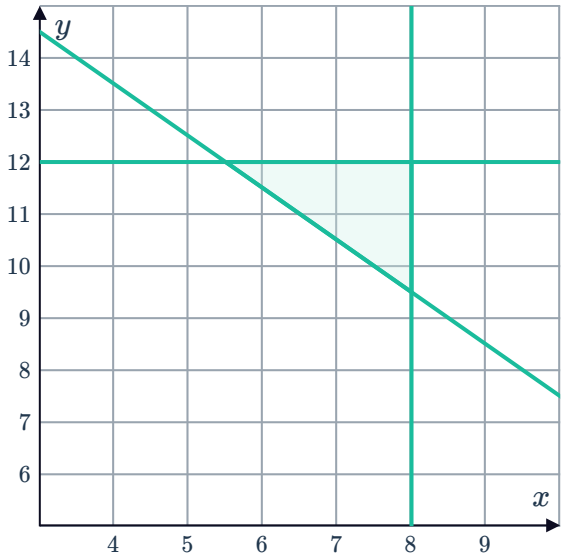
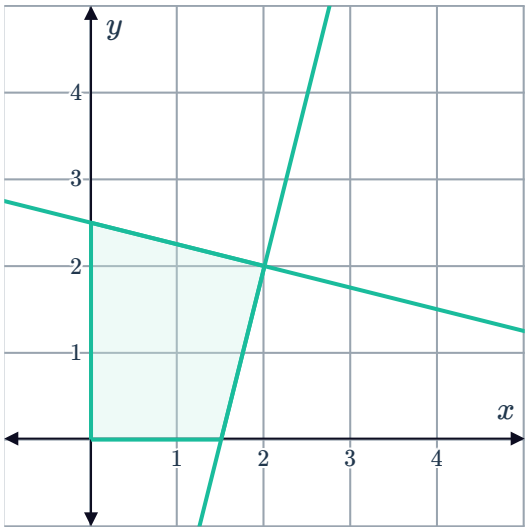
Integer solutions in feasible regions

1 Given the following graphs, list the **integer** coordinate pairs that satisfy the feasible region which is shaded:

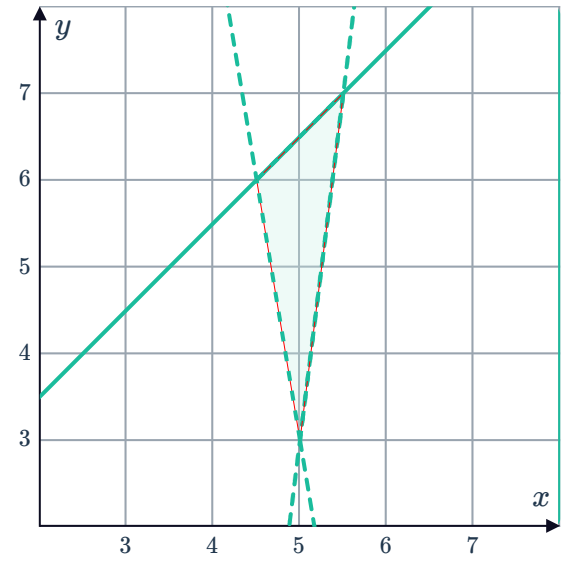
a



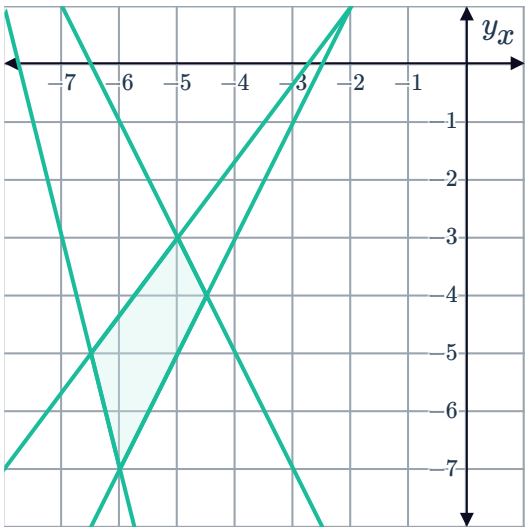
b



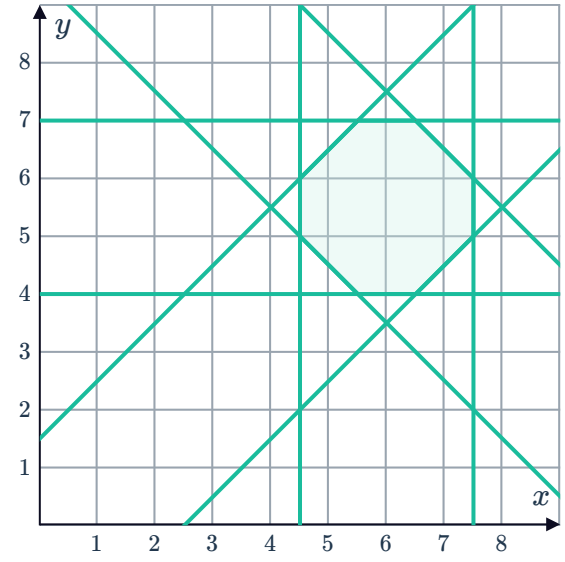
c



d



e



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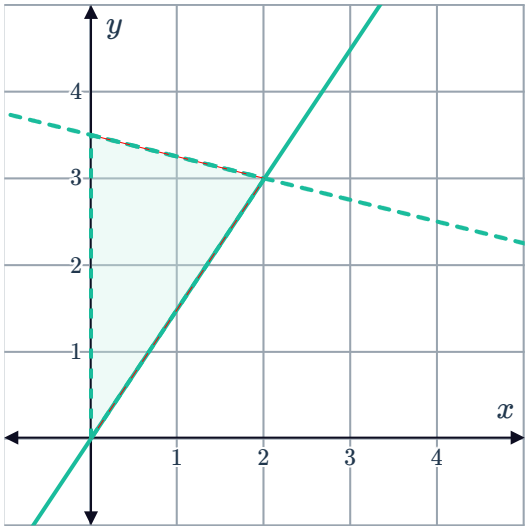
2 For each of the following objective functions:

i List the integer coordinate pairs that satisfy the feasible region on the corresponding graph.

ii Find the integer coordinate pair that maximises the objective function.

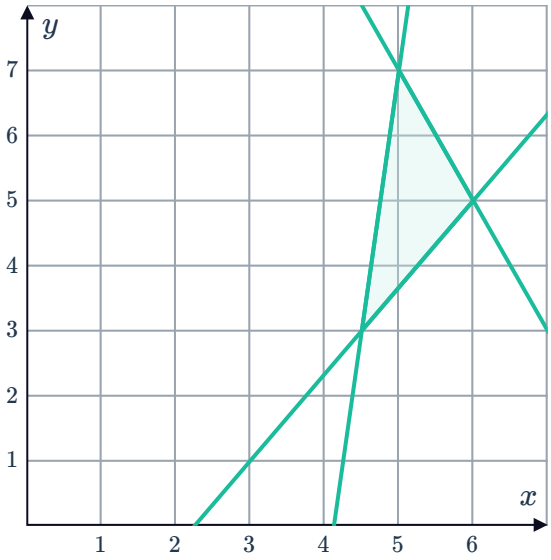
a The objective function for the graph is:

$$z = -\frac{x}{3} + \frac{y}{3} + 7$$



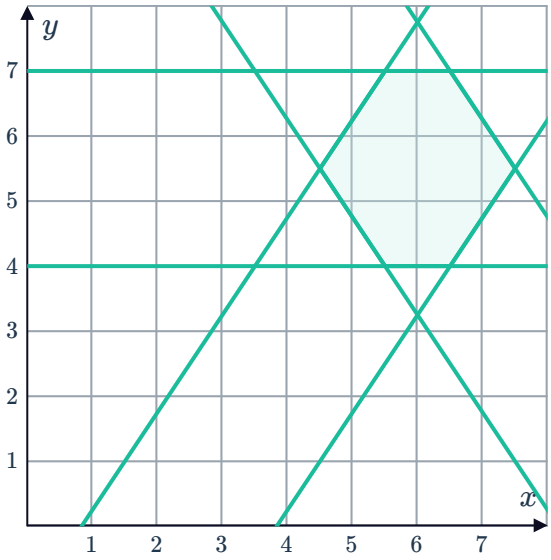
b The objective function for the graph is:

$$z = \frac{x}{2} - \frac{3y}{2} + 30$$



c The objective function for the graph is:

$$z = -\frac{8x}{3} + \frac{7y}{3} + 7$$



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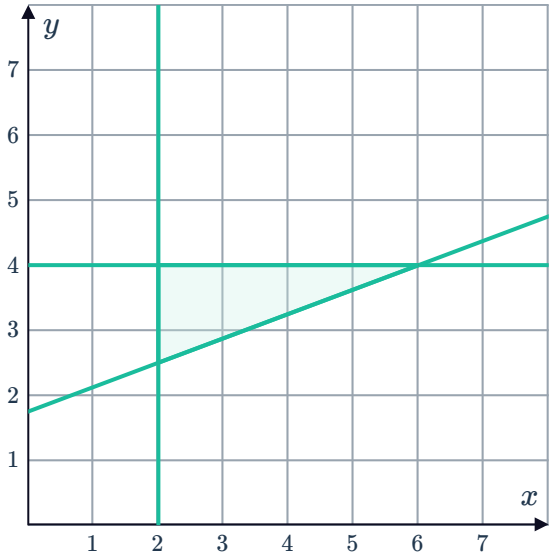
3 For each of the following objective functions:

i List the integer coordinate pairs that satisfy the feasible region on the corresponding graph.

ii Find the integer coordinate pair that minimises the objective function.

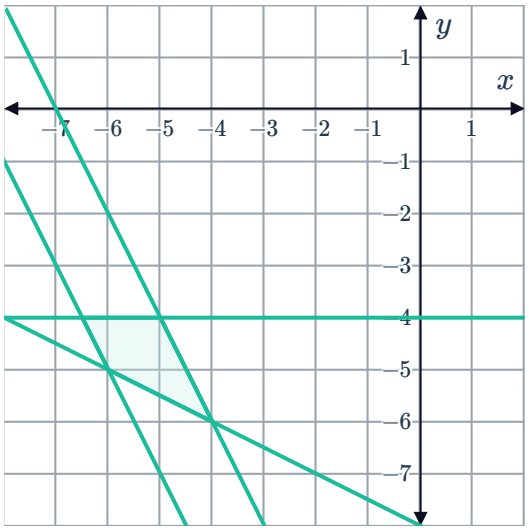
a The objective function for the graph is:

$z = x + 2y + 15.$



b The objective function for the graph is:

$z = -\frac{3x}{4} - \frac{3y}{4} + 15.$



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Integer solutions in linear programming applications

4 A tutoring company offers private tuition for English and Mathematics. An English student takes one tuition session which is broken up into two sections that cover Module A and Module B. The same goes for a Mathematics student, but with a different amount of hours covering each module. The company decides to set a maximum number of hours dedicated to each module according to its relevance in the syllabus.

The maximum hours and set by the company and the break up of a tuition session is shown in the table:

	Module A	Module B
English session	3 hours	1 hour
Mathematics session	2 hours	3.5 hours
Max time per week	24 hours	14 hours

Let x represent the number of students that take English per week, and let y represent the number of students that take Mathematics per week.

a Explain why x and y must be integer values.

b Construct the set of four constraint inequalities.

c Graph the region defined by the constraint inequalities.

d If English tuition is charged at $\$50$ per hour and Mathematics is charged at $\$40$ per hour, write the objective function that models the weekly profit, P.

e The set of integer solutions that could minimise the objective function is listed in the table:

Complete the table to determine the profit corresponding to each possible coordinate pair.

Possible solutions	Profit ($)
$(0, 4)$	
$(7, 0)$	
$(6, 2)$	

f Hence, state the maximum weekly profit.

g State the number of English students and Mathematics students required for the company to achieve the maximum profit.

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5 A bakery sells gourmet cakes and muffins. The two most expensive ingredients are vanilla beans and truffles. The amounts of each ingredient required to make each baked good is given in the table. The bakery must use at least the stated amounts in the last row of each ingredient before expiry:

	Vanilla beans	Truffles
Amount required for one cake	8 g	6 g
Amount required for one muffin	2 g	4 g
Amount of each ingredient that must be used	24 g	36 g

Let x be the number of cakes made and let y be the number of muffins made.

a Construct the set of four constraint inequalities.

b Graph the region defined by the constraint inequalities.

c The cost of vanilla beans is $\$0.60$ per gram, and the cost of truffles is $\$3.50$ per gram. Write the objective function C that describes the cost of the ingredients.

d Write the set of integer solutions that could minimise the cost function.

e Determine the cost corresponding to each of the possible integer solutions in the feasible region.

f Hence, state the minimum cost for ingredients.

g State the number of cakes and muffins that should be made to minimise the cost of ingredients.

6 Beth wants to plant carrots and parsnips in her veggie patch. The planting space and potting mix required for each plant are given below:

	Planting space	Potting mix
Carrots	175 cm^2	0.25 cm^2
Parsnips	250 cm^2	0.8 cm^2
Maximum amount available	5250 cm^2	8 cm^2

Let x be the number of carrots and let y be the number of parsnips.

a Construct the set of four constraint inequalities.

b Graph the region defined by the constraint inequalities.

c Beth wants to plant as many carrots or parsnips as she can. Write the objective function, P, that describes the number of carrots and parsnips.

d Write the set of integer solutions that could maximise the objective function, P.

e Determine the maximum number of plants that can fit in Beth's veggie patch.

f State the number of carrots and parsnips Beth should plant if she wants to fit as many plants as possible into her veggie patch.

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7 Bill wants to grow tomato plants and strawberry plants. The amount of water and fertiliser required for each plant per week is given in the table below. To save on resources, the amount of water and fertiliser used must be within the maximum weekly amount:

	Water	Fertilizer
Tomato plant	15 ml	20 ml
Strawberry plant	10 ml	15 ml
Maximum weekly amount	1000 ml	600 ml

Let x be the number of tomato plants and let y be the number of strawberry plants.

a Bill wants at least 10 tomato plants and 10 strawberry plants in his garden. Construct the set of four constraint inequalities.

b Graph the region defined by the constraint inequalities.

c A tomato plant yields 10 tomatoes, and each tomato sells for $\$0.50$. A strawberry plant yields 20 strawberries, and each strawberry sells for $\$0.10$. Write the objective function, V , that models the value of the tomatoes and strawberries in dollars.

d List the set of integer solutions from the feasible region that could maximise the objective function.

e Determine the value of V corresponding to each coordinate pair from the feasible region.

f Hence, state the maximum value for the garden.

g State the number of each type of plant that Bill should grow to maximise the value of his garden.

8 A health foods startup wants to sell two new products, pea protein powder and pea protein pancake mix. The amount of protein and carbohydrates for each product is given below:

The startup wants to use at least 300 g of pea protein isolate in their first test batch, and at most 350 g of carbohydrates worth of flavouring. They also want to make at least 9 of each product.

	Protein	Carbohydrates
Protein powder	25 g	12 g
Pancake mix	15 g	15 g

Let x be the number of protein powder sachets and y be the number of pancake mix packets.

a Construct the set of four constraint inequalities.

b Use your CAS calculator to graph the regions as defined by the constraint inequalities.

c The startup plans to sell the protein powder for $\$15$ per sachet and the pancake mix for $\$12$. Write the objective function, V, that models the value of their first batch in dollars.

d List the set of all integer solutions in the feasible region that could minimise the objective function.

e Determine the value, V, corresponding to each integer solution in the feasible region

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f State the maximum value for the first batch.

g State the number of protein powder sachets and packets of pancake mix needed to maximise their first sales.

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